

Seonghyun Chon

RAW PERCEPTION / AUTONOMOUS SYSTEMS / ROBOTICS DATA

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EDUCATION & SCHOLARSHIP

Hanyang University

Mar 2021 - Present

B.S. in progress, Automotive Engineering

Double major: Automotive Software

LG Electronics Industry-Academia Scholar

Sep 2025 - Present

RESEARCH EXPERIENCE

Undergraduate Research Intern • IRCV Lab

Feb 2026 - Present

Intelligent Robotics and Computer Vision Laboratory, Hanyang University

Advisor: Soonmin Hwang

RIFT & DriveRAW • RAW-domain object detection

- Led experiments for a learned RAW-to-detector frontend; integrated multi-dataset baselines, controlled ablations and cross-domain diagnostics in PyTorch/MMDetection.
- Contributed paired low-light RAW/RGB driving-data collection and annotation, with reproducible preprocessing, training and evaluation tooling.

ISP-stage study • Research manuscript

- Designed and ran controlled experiments across RAW-to-sRGB processing stages to separate image-signal effects from pretraining-dependent effects.
- Developed Bayer-aware preprocessing, annotation alignment, experiment queues and exposure/bit-depth diagnostics across multiple camera datasets.

Additional research prototypes

- **RAW-assisted auto-labeling:** combined consistency across image renders, weighted box fusion and sensor-quality cues to generate and evaluate pseudo-labels. [Code]
- **RAW uncertainty distillation:** prototyped RAW-teacher supervision for RGB-only 3D perception using synthetic sensor noise and physical reliability priors.

LEADERSHIP & COMPETITION RESULTS

2nd President • MIRU (MiRuAE)

Sep - Dec 2025

Grand Prize • 1/10-scale division

Jul 2025

4th International University Student EV Self-Driving Competition

Tied 9th place • F1TENTH Korea Championship

Nov 2025

4th F1TENTH Korea Championship, ICCAS 2025

SELECTED TALK

ISP and low-light structure from motion (Dark3R)

Aug 14, 2026

IRCV Lab research seminar

TECHNICAL SKILLS

Programming: Python, C++, C **Learning:** PyTorch, MMDetection, YOLO

Robotics: ROS 2, Gazebo, SLAM, camera/LiDAR calibration, iLQR, DAGger/PPO

Data & systems: RAW imaging, sensor synchronization, nuScenes, rosbag, FastAPI, xv6

SENSOR, DATASET & FIELD INFRASTRUCTURE

Data Machine • Full-stack sensor platform

2026

- Authored C++ Spinnaker/ROS 2 drivers and typed GenICam controls for 14 RGB and 2 thermal cameras; integrated LiDAR/GNSS, device tuning, ForcelP/GPIO/PTP, calibration and the operator UI.
- Built a C++ serialized-message ring-buffer recorder with adaptive QoS, pre/post event windows and background rosbag writing; unified sensor launch, previews and recording-integrity diagnostics.
- Integrated CUDA/NPP Bayer demosaicing and nvJPEG in the **FLIR sensor stack**, with exposure-time correction and PTP verification. Demonstrated 30 Hz camera recording without observed drops in the September 19, 2026 test.

T-Car • Multi-sensor data and perception

2026

- Developed online camera-LiDAR calibration for seven cameras, jointly refining lens/windshield models, rolling shutter, timing and extrinsics through KLT bundle adjustment, LoFTR/PnP and joint optimization; exported dataset calibration.
- Contributed synchronization-aware ROS-to-nuScenes conversion, dropped-frame handling, camera rectification, INS/radar exports and resumable batch ingestion. [\[Code\]](#)
- Developed two-stage YOLO traffic-light detection, seven-state classification and 2D object-detection tooling; built a FastAPI map/GPS route-collection platform with versioned routes.

AUTONOMOUS DRIVING & RACING

Team ChaChaPing • EV competition autonomy

Feb - Jul 2025

- Implemented YOLO segmentation for lane detection, polynomial lane fitting and EMA center tracking; integrated Ackermann control and mission behavior.

ICCAS F1TENTH competition stack

Sep - Nov 2025

- Contributed adaptive lane/waypoint tracking, speed- and curvature-dependent lookahead, track-map integration and IMU/odometry interfaces to Team 폭주(暴走)'s ROS 2 racing stack.

HYU Formula Student • Perception & SLAM

2026

- Developed and integrated camera/LiDAR cone perception, g2o GraphSLAM, and local/global planning and control for Formula Student Korea 2026 preparation.
- Extended EUFS/Gazebo with sensor timing/distortion models, stereo alignment and closed-loop validation; integrated racing-line and velocity-profile optimization.

F1TENTH E2E • Custom simulator & learning

Sep 2026

- Built a GPU-batched PyTorch racing simulator with vehicle physics, sensor models and a track editor; calibrated it from 22 real recordings and integrated DAgger/PPO, LiDAR-to-local-plan policies and iLQR control for simulation research.

EMBEDDED AI & SYSTEMS COURSEWORK

Embedded-TARS • Voice-guided mobile robot

May - Jun 2025

- Implemented Whisper-to-Phi-4-mini/BitNet voice-command processing, constrained JSON navigation/manual commands and gTTS responses; contributed camera YOLO lane/object inference.

Operating systems • xv6

Apr - Jun 2026

- Implemented a CFS-style scheduler with virtual runtime, a red-black tree and nice values; clone/join shared-address-space threads and a user library; lazy file-backed mmap/munmap.